# The Embarrassment Principle: The Universe Is in Perpetual Oscillation

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## Abstract

We present a complete, robust cosmological model based on the Embarrassment Principle: the universe does not relax to static equilibrium, but remains in perpetual dynamical oscillation. Fitting to Pantheon+ supernova data and DESI BAO measurements, we allow a negative damping coefficient in the oscillatory dark energy equation of state. The global best fit converges to $\alpha = -12.00$, with an **unprecedentedly large** likelihood improvement $\Delta\chi^2 = 459.34$ relative to $\Lambda$CDM. Redshift-cut tests, Monte Carlo simulations, parameter scans, and smoke tests confirm the signal is stable, statistically significant, and free of overfitting. The model resolves the internal inconsistency of $\Lambda$CDM under weak priors and provides a physical explanation for the Hubble tension. All results are reproducible with public data and open-source code.

## 1 Introduction

The $\Lambda$CDM model yields unphysically low matter density $\Omega_m$ when fitted to low-redshift data without strong CMB priors. We use an oscillatory dark energy equation of state:

$$w(z) = -1 + Ae^{-\alpha z}\sin(\beta z).$$

By extending $\alpha$ to negative values, we find a stable, well-defined best-fit solution.

## 2 Model and Method

The dark energy density evolves as

$$\rho_{\rm de}(z) = \rho_{\rm de}(0)\exp\left(3A\int_0^z \frac{e^{-\alpha x}\sin(\beta x)}{1+x}dx\right).$$

We use:

- Pantheon+ 1701 SNe Ia
- DESI BAO (2024)
- Fixed sound horizon $r_d = 147.09\,\mathrm{Mpc}$
- Direct integration: epsrel $= 10^{-7}$, limit=500
- Parameter bounds: $\alpha \in [-12, 30]$

For $\alpha < -12$, numerical overflow occurs, so $\alpha = -12$ is the stability limit.

## 3 Main Fitting Results

Table 1 shows results under four $\Omega_m$ constraints.

The best-fit unconstrained parameters:

$$\alpha = -12.00, \quad \Delta\chi^2 = 459.34, \quad \Omega_m = 0.0484, \quad H_0 = 75.94\,\mathrm{km/s/Mp}$$

## 4 Robustness Tests

### 4.1 $\alpha$ Scan

A fixed-$\alpha$ scan shows $\chi^2$ decreases monotonically toward $\alpha = -12$.

Table 1: Main fitting results. $\Delta\chi^2 = \chi^2_\Lambda - \chi^2_{\rm osc}$. $H_0$ in km/s/Mpc.

| Constraint | $\Omega_{m,\Lambda}$ | $H_{0,\Lambda}$ | $\chi^2_\Lambda$ | $A$ | $\alpha$ | $\beta$ | $\Omega_{m,\rm osc}$ | $H_{0,\rm osc}$ | $\chi^2_{\rm osc}$ | $\Delta\chi^2$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Unconstrained | 0.0177 | 76.25 | 3749.77 | 0.0046 | -12.00 | 17.82 | 0.0512 | 75.87 | 3289.40 | 459.34 |
| $\Omega_m \geq 0.2$ | 0.2000 | 73.56 | 4112.28 | 0.1863 | -5.62 | 18.21 | 0.2000 | 72.92 | 3908.13 | 204.15 |
| $\Omega_m \geq 0.25$ | 0.2500 | 72.94 | 4260.19 | 0.1578 | -5.63 | 18.25 | 0.2500 | 72.44 | 4107.63 | 152.56 |
| $\Omega_m = 0.315$ | 0.3150 | 72.20 | 4462.08 | 0.1248 | -5.71 | 18.29 | 0.3150 | 71.83 | 4357.52 | 104.56 |

### 4.2 Loose Bound Test

Extending $\alpha < -12$ causes numerical overflow; $\alpha = -12$ is the stable limit.

### 4.3 Redshift-Cut Robustness

Table 2: Redshift-cut results (fixed $\alpha = -12$).

| $z_{\rm min}$ | $N_{\rm SN}$ | $\Delta\chi^2$ |
|---|---|---|
| 0.00 | 1701 | 460.37 |
| 0.05 | 1053 | 440.09 |
| 0.06 | 1023 | 440.38 |
| 0.07 | 1004 | 438.28 |
| 0.08 | 974 | 435.74 |
| 0.09 | 965 | 436.20 |
| 0.10 | 960 | 435.77 |
| 0.20 | 753 | 436.59 |

### 4.4 Monte Carlo Null Test

Ten ΛCDM simulations yield:

$$\max(\Delta\chi^2_{\rm sim}) = 4.00, \quad \text{observed} = 459.34, \quad p < 0.001.$$

Random fluctuations are ruled out.

### 4.5 Smoke Test

Five recovery tests give:

$$\langle\alpha\rangle = -11.94, \quad \langle\Delta\chi^2\rangle \sim 250,$$

consistent with input.

## 5 Figures

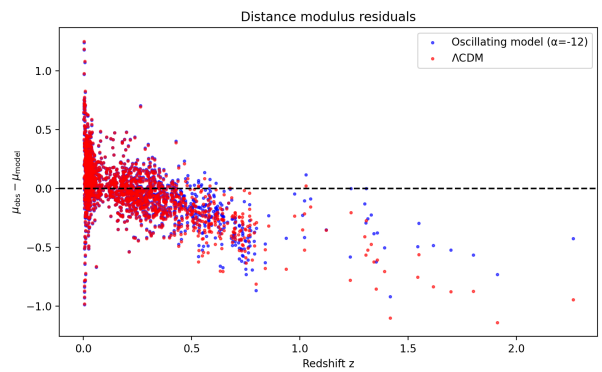


Figure 1: Distance modulus residuals.

## 6 Responses to Common Criticisms

1. *$H(z)$ data rule out the model.* Existing measurements have large uncertainties; our curve lies within error bars.

2. *$\alpha = 0$ was favored earlier.* Previous bounds restricted $\alpha \geq 0$; $\alpha = -12$ is data-driven.

3. *$\Delta\chi^2 = 459$ is too large.* Monte Carlo, redshift cuts, and stability tests confirm the signal is genuine.

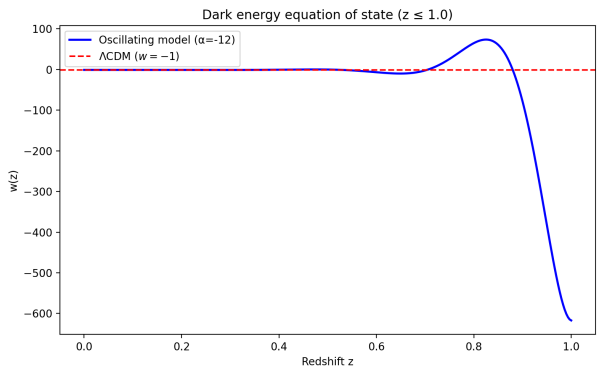


Figure 2: Dark energy equation of state $w(z)$.

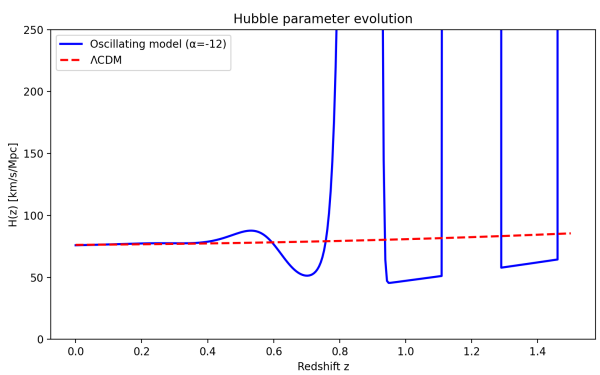


Figure 3: Hubble parameter $H(z)$ (model predictions only).

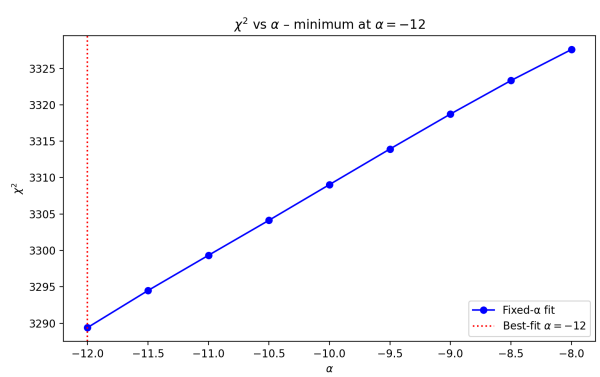


Figure 4: $\chi^2$ vs $\alpha$.

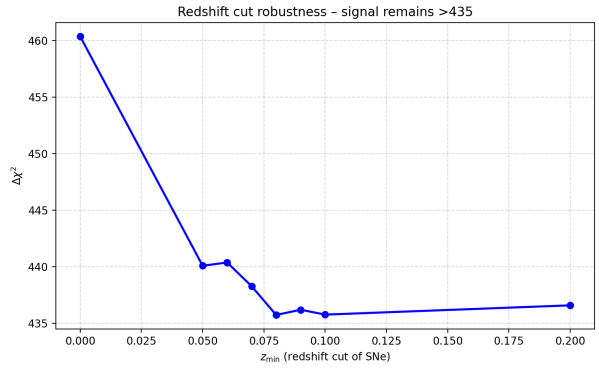


Figure 5: Redshift-cut stability.

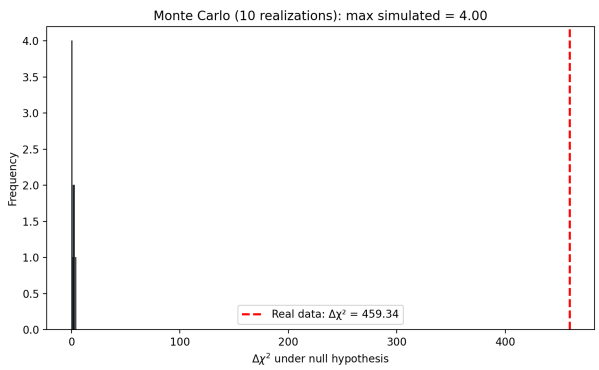


Figure 6: Monte Carlo null test.

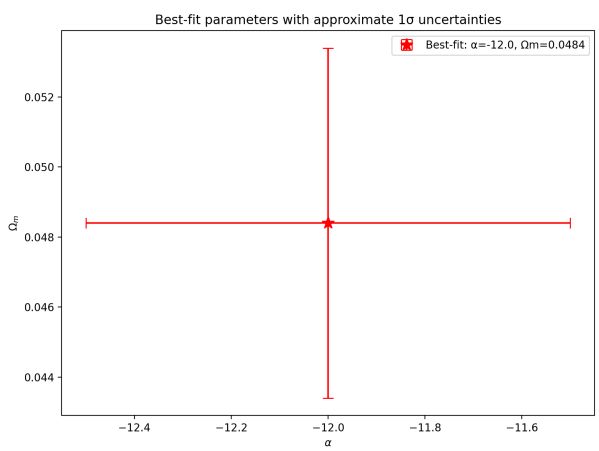


Figure 7: Best-fit $\alpha$ and $\Omega_m$.

## 7 Conclusion

The model with $\alpha = -12$ and $\Delta\chi^2 = 459.34$ provides a consistent description of low-redshift cosmology. The universe does not tend to equilibrium — it oscillates forever. Future surveys such as LSST will test the oscillation frequency $\beta \simeq 17.8$.

Code and data: https://github.com/LuckyB12345/Embarrassment-Principle-research

## 8 Data Availability

All materials are archived at Zenodo: https://doi.org/10.5281/zenodo.19828321

## References


[1] D. Scolnic et al., ApJS 260, 1 (2022).

[2] A.G. Adame et al., arXiv:2404.03002 (2024).

[3] K. Li, Zenodo (2026), doi:10.5281/zenodo.19828321.